# A CONSISTENT INDEPENDENCE TEST

## VIA PROJECTED MUTUAL INFORMATION

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# Problem of Interests

Suppose $\mathbf{x} = (X_1, \ldots, X_p)^\top \in \mathbb{R}^p$ and $\mathbf{y} = (Y_1, \ldots, Y_q)^\top \in \mathbb{R}^q$ are two random vectors. Our primary interest is to test statistical independence of these two random vectors. In symbols, we aim to test

$$H_0 : \mathbf{x} \perp\!\!\!\perp \mathbf{y} \quad \text{versus} \quad H_1 : \text{ otherwise.}$$

# Applications: Independence Test

- In biostatistics, test relations between certain phenotypes and genotypes.
- In social science, test potential associations between psychological and physiological characteristics.
- ....

# Literature review: Pearson correlation.

Pearson correlation measures the linear relationship:

$$\rho_{X,Y} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Under the assumption that $X$ and $Y$ are jointly normal,

$$\rho_{X,Y} = 0 \iff X \perp\!\!\!\perp Y$$

# Literature review: Rank-based measures.

Let $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ be observations from $(X, Y)$. Then Kendall's rank coefficient is

$$\tau = \frac{2}{n(n-1)} \sum_{i<j} \mathrm{sgn}(X_i - X_j)\mathrm{sgn}(Y_i - Y_j)$$

It assess the monotonic association between $X$ and $Y$.

# Literature review: Distance Correlation.

The distance correlation [Székely et al., 2007] measures the difference between the joint characteristic function $\phi_{\mathbf{x},\mathbf{y}}$ and the product of marginal characteristic functions $\phi_{\mathbf{x}}$, $\phi_{\mathbf{y}}$:

$$\int_{\mathbb{R}^{p+q}} |\phi_{\mathbf{x},\mathbf{y}}(t,s) - \phi_{\mathbf{x}}(t)\phi_{\mathbf{y}}(s)|^2 w(t,s) dtds$$

It characterizes dependence between two random variables of arbitrary dimension, provided the moments exist.

# Literature review: Projection Correlation.

Zhu et al. [2017] extended the Hoeffding coefficient [Hoeffding, 1948] via projection pursuit and proposed the projection correlation. Essentially, it transformed the testing independence problem into testing whether

$$\int\int\int \mathrm{cov}^2\{I(\alpha^\top \mathbf{x} \leq u), I(\beta^\top \mathbf{y} \leq v)\} dF_{U,V}(u,v) d\alpha d\beta$$

equals to zero or not.

# Introduction: Mutual Information

- For random vectors $\mathbf{x}$ and $\mathbf{y}$, the mutual information is defined as

$$I(\mathbf{x}, \mathbf{y}) = \mathrm{E}\left\{\log \frac{f_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{f_{\mathbf{x}}(\mathbf{x}) f_{\mathbf{y}}(\mathbf{y})}\right\},$$

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- It is non-negative, equals to zero if and only if $\mathbf{x}$ and $\mathbf{y}$ are statistically independent.
- It is invariant with respect to reparametrizations. Let $\mathbf{x}' = T_1(\mathbf{x})$ and $\mathbf{y}' = T_2(\mathbf{y})$ be smooth and uniquely invertible transformations of $\mathbf{x}$ and $\mathbf{y}$. Then $I(\mathbf{x}', \mathbf{y}') = I(\mathbf{x}, \mathbf{y})$.

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- Berrett and Samworth [2019] propose a MINT test, which estimates the mutual information via efficient entropy functions. But it can suffer power loss under high dimensions.

# Toy Example: High Dimensional Effect

**Table 1:** The empirical size and power of Berrett and Samworth [2019] when testing $H_0: \ Y \perp\!\!\!\perp \mathbf{x}$, where $\mathbf{x} = (X_1, \ldots, X_p)^\mathsf{T}$ follows a multivariate normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$ and $Y = c_0 X_1 + \varepsilon$. We fix the sample size $n = 50$ and evaluate different dimensions of the covariates $\mathbf{x}$ with $p = 2, 4, 6, 8, 10$. We investigate different significance levels with $\alpha = 0.05, 0.1$

| $c_0$ | $\alpha$ | $p = 2$ | $p = 4$ | $p = 6$ | $p = 8$ | $p = 10$ |
|---|---|---|---|---|---|---|
| 0 | 0.05 | 0.046 | 0.040 | 0.056 | 0.056 | 0.043 |
| | 0.10 | 0.071 | 0.100 | 0.084 | 0.079 | 0.090 |
| 1 | 0.05 | 0.958 | 0.773 | 0.584 | 0.548 | 0.476 |
| | 0.10 | 0.978 | 0.847 | 0.732 | 0.672 | 0.602 |

# Methodology: Tackling High Dimensions

- We note that

$$I(\mathbf{x}, \mathbf{y}) = \mathrm{E}\left\{\log \frac{f_{\mathbf{x},\mathbf{y}}(\mathbf{x}, \mathbf{y})}{f_{\mathbf{x}}(\mathbf{x}) f_{\mathbf{y}}(\mathbf{y})}\right\} = \mathrm{E}\left\{\log \frac{f_{\mathbf{y}|\mathbf{x}}(\mathbf{y} \mid \mathbf{x})}{f_{\mathbf{y}}(\mathbf{y})}\right\},$$

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- The estimation of $I(\mathbf{x}, \mathbf{y})$ can be replaced with $I(\boldsymbol{\beta}_0^\top \mathbf{x}, \mathbf{y})$, which is a low dimensional problem

# Methodology: Tackling High Dimensions

- In fact, if such $\boldsymbol{\beta}_0$ exists,

$$\boldsymbol{\beta}_0 = \arg\max_{\boldsymbol{\beta}} \mathrm{E}\left\{ \log \frac{f_{\mathbf{y}|\boldsymbol{\beta}^\top \mathbf{x}}(\mathbf{y} \mid \boldsymbol{\beta}^\top \mathbf{x})}{f_{\mathbf{y}}(\mathbf{y})} \right\},$$

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- Therefore, we can estimate $\boldsymbol{\beta}_0$ by maximizing the empirical mutual information $I(\boldsymbol{\beta}^\top \mathbf{x}, \mathbf{y})$.
- Under the alternative, i.e., $\mathbf{y}$ depends on $\mathbf{x}$, such $\boldsymbol{\beta}_0$ exists. However, when $\mathbf{x}$ and $\mathbf{y}$ are independent, $\boldsymbol{\beta}_0$ is not identifiable.

# Methodology: Identifiability of $\beta_0$

- Under the null hypothesis, $\sup_{\boldsymbol{\beta}} I(\boldsymbol{\beta}^\top \mathbf{x}, \mathbf{y}) = 0$ for any $\boldsymbol{\beta}$.

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- Thus we can't directly identify $\boldsymbol{\beta}_0$ by maximizing the mutual information.
- We may incorporate a penalty term to force the maximum to be attained at a pre-specified value $\boldsymbol{\beta}^*$ under the null hypothesis.
- Specifically,

$$\boldsymbol{\beta}_0 = \arg\max \left\{ I(\boldsymbol{\beta}^\top \mathbf{x}, \mathbf{y}) - c\mathbb{I}(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*) \right\},$$

# Methodology: Penalized Mutual Information

In practice, we fix the dimension of $\widehat{\boldsymbol{\beta}}$ to be 1 and call it the maximum unit direction. Specifically, define

$$\widehat{\boldsymbol{\beta}} = \underset{\|\boldsymbol{\beta}\|=1}{\arg\max} \left\{ U_n(\boldsymbol{\beta}) - c_n I(\boldsymbol{\beta} \neq \boldsymbol{\beta}^*) \right\},$$

where $\boldsymbol{\beta}^* \in \mathbb{R}^p$ and $\|\boldsymbol{\beta}^*\| = 1$. $c_n$ is a sequence of positive real numbers decreasing to zero at an appropriate rate. $U_n(\boldsymbol{\beta})$ is the empirical version of $I(\mathbf{x}^\top \boldsymbol{\beta}, Y)$, which is defined as

$$U_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^{n} \log \left\{ \frac{\widehat{f}_{Y|\mathbf{x}^\top \boldsymbol{\beta}}(Y_i \mid \mathbf{x}_i^\top \boldsymbol{\beta})}{\widehat{f}_Y(Y_i)} \right\},$$

# Methodology: Penalized Mutual Information

$\widehat{f}_{Y|\mathbf{x}^\top\boldsymbol{\beta}}(Y_i \mid \mathbf{x}_i^\top\boldsymbol{\beta})$ and $\widehat{f}_Y(Y_i)$ are the leave-one-out kernel density estimators. Specifically, they are defined as

$$\widehat{f}_{Y|\mathbf{x}^\top\boldsymbol{\beta}}(Y_i \mid \mathbf{x}_i^\top\boldsymbol{\beta}) = \frac{\sum_{j\neq i}^n K_{h_x}(\mathbf{x}_i^\top\boldsymbol{\beta}-\mathbf{x}_j^\top\boldsymbol{\beta})K_{h_y}(Y_i-Y_j)}{\sum_{j\neq i}^n K_{h_x}(\mathbf{x}_i^\top\boldsymbol{\beta}-\mathbf{x}_j^\top\boldsymbol{\beta})},$$

$$\widehat{f}_Y(Y_i) = \frac{1}{n-1}\sum_{j\neq i}^n K_{h_y}(Y_i - Y_j).$$

Here $K_{h_x}(\cdot) = K(\cdot/h_x)/h_x, K_{h_y}(\cdot) = K(\cdot/h_y)/h_y$, $K(\cdot)$ is a kernel function, while $h_x$ and $h_y$ are bandwidths.

# Computation of $\widehat{\beta}$

1. Input error control $\epsilon$, maximum iteration $M$. Initialize $\boldsymbol{\beta}$ with a root-$n$ consistent estimator.
2. While $1 \leq m \leq M$
   2.1 Compute the current gradient matrix $\mathbf{G}$ of $U_n(\boldsymbol{\beta})$ at $\boldsymbol{\beta}_{m-1}$.
   2.2 Compute the skew-symmetric matrix $\mathbf{A} = \mathbf{G}\boldsymbol{\beta}_{m-1}^{\mathsf{T}} - \boldsymbol{\beta}_{m-1}\mathbf{G}^{\mathsf{T}}$.
   2.3 Compute the Barzilai-Borwein step size $\tau$ with nonmonotone line search.
   2.4 Set $\boldsymbol{\beta}_m = \boldsymbol{\beta}_{m-1}(\tau)$, where $\boldsymbol{\beta}_{m-1}(\tau)$ is given by

$$\boldsymbol{\beta}(\tau) = \left(\mathbf{I} + \frac{\tau}{2}\mathbf{A}\right)^{-1}\left(\mathbf{I} - \frac{\tau}{2}\mathbf{A}\right)\boldsymbol{\beta}$$

   2.5 Break if $\|\boldsymbol{\beta}_m - \boldsymbol{\beta}_{m-1}\|_2 \leq \epsilon$.
3. Output $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}_m$ if $U_n(\boldsymbol{\beta}_m) - c_n I(\boldsymbol{\beta}_m \neq \boldsymbol{\beta}^*) > U_n(\boldsymbol{\beta}^*)$ and $\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ otherwise.

# Asymptotic Properties of $\widehat{\beta}$

- Under the null hypothesis, $\sup_{\|\beta\|=1} U_n(\beta) = O_p(n^{-1}h_x^{-1}h_y^{-1}\log n)$. Furthermore, when $c_n n h_x h_y / \log n \to \infty$, we have $\mathrm{P}(\widehat{\beta} = \beta^*) \to 1$.

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- Under the alternative hypothesis, when $c_n = o(n^{-1/2})$,

$$n^{1/2}\mathbf{B}_1(\widehat{\beta} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{B}_2\right),$$

where $\mathbf{B}_1$ and $\mathbf{B}_2$ are given in the appendix.

# Methodology: The Test Statistic

- After obtaining $\widehat{\boldsymbol{\beta}}$, the test statistic is constructed by applying the mutual information as

$$T_n = U_n(\widehat{\boldsymbol{\beta}}) = \frac{1}{n}\sum_{i=1}^{n} \log\left\{\frac{\widehat{f}_{\mathbf{y}|\boldsymbol{\beta}^\top\mathbf{x}}(\mathbf{y}_i \mid \widehat{\boldsymbol{\beta}}^\top \mathbf{x}_i)}{\widehat{f}_{\mathbf{y}}(\mathbf{y}_i)}\right\}.$$

- Under the null hypothesis, $T_n$ converges to $I({\boldsymbol{\beta}^*}^\top\mathbf{x}, \mathbf{y}) = 0$ with probability tending to 1.
- While under the alternative hypothesis, $T_n$ converges in probability to $I(\mathbf{x}, \mathbf{y}) > 0$
- Larger $T_n$ provides more evidence against the null hypothesis.

# Asymptotic Properties of the Test Statistic

- Under the null hypothesis, with probability tending to 1,

$$
\begin{aligned}
& nh_x^{1/2} h_y^{1/2} \Bigg[ T_n + \frac{\nu_2^2}{2nh_x h_y} E\left\{ f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x}, Y}(\boldsymbol{\beta}^{*\top}\mathbf{x}, Y) \right\} \\
& \qquad - \frac{\nu_2}{2nh_x} E\left\{ f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x}}(\boldsymbol{\beta}^{*\top}\mathbf{x}) \right\} - \frac{\nu_2}{2nh_y} E\left\{ f_Y^{-1}(Y) \right\} \Bigg] \\
\xrightarrow{d} \quad & \mathcal{N}\left[ 0, 2^{-1}\nu_2^2 E\left\{ f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x}, Y}(\boldsymbol{\beta}^{*\top}\mathbf{x}, Y) \right\} \right],
\end{aligned}
$$

where $\nu_2 \stackrel{\text{def}}{=} \int K^2(u)du$ and $K(\cdot)$ is the kernel function

# Asymptotic Properties of the Test Statistic

- Under the global alternative hypothesis, $I(\mathbf{x}, \mathbf{y}) > 0$, when $c_n = o(n^{-1/2})$,

$$n^{1/2}\left\{T_n - I(\mathbf{x}, Y)\right\} \xrightarrow{d} \mathcal{N}\left(0, \mathrm{Var}\left[\log\left\{\frac{f_{Y|\boldsymbol{\beta}_0^\top \mathbf{x}}(Y \mid \boldsymbol{\beta}_0^\top \mathbf{x})}{f_Y(Y)}\right\}\right]\right).$$

# Asymptotic Properties of the Test Statistic

- Under the local alternative hypothesis, $f_{\mathbf{y}|\boldsymbol{\beta}^\top\mathbf{x}}(\mathbf{y} \mid \boldsymbol{\beta}^\top\mathbf{x}) = f_{\mathbf{y}}(\mathbf{y})\left\{1 + \alpha_n g(\boldsymbol{\beta}^\top\mathbf{x}, \mathbf{y})\right\}$ for all $\boldsymbol{\beta}$, when $\alpha_n = n^{-1/2}h_x^{-1/4}h_y^{-1/4}$, with probability tending to 1, we have

$$
\begin{aligned}
& nh_x^{1/2}h_y^{1/2}\Bigg[T_n + \frac{\nu_2^2}{2nh_xh_y}E\left\{f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x},Y}(\boldsymbol{\beta}^{*\top}\mathbf{x}, Y)\right\} \\
& \qquad - \frac{\nu_2}{2nh_x}E\left\{f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x}}(\boldsymbol{\beta}^{*\top}\mathbf{x})\right\} - \frac{\nu_2}{2nh_y}E\left\{f_Y^{-1}(Y)\right\}\Bigg] \\
\xrightarrow{d} \;& \mathcal{N}\left[2^{-1}\Delta, 2^{-1}\nu_2^2 E\left\{f^{-1}_{\boldsymbol{\beta}^{*\top}\mathbf{x},Y}(\boldsymbol{\beta}^{*\top}\mathbf{x}, Y)\right\}\right],
\end{aligned}
$$

- The local smoothing based tests can generally detect local alternatives converging to the null at the rate $n^{-1/2}h_x^{-p/4}h_y^{-1/4}$, which is slower than ours tests.

# Extension to Arbitrary Dimensions

For arbitrary dimensions of $\mathbf{x}$ and $\mathbf{y}$, we can perform projections on both random vectors. Specifically, define

$$(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = \underset{\|\boldsymbol{\alpha}\|=1, \|\boldsymbol{\beta}\|=1}{\arg\max} \left[ V_n(\boldsymbol{\alpha}, \boldsymbol{\beta}) - c_n I \left\{ (\boldsymbol{\alpha}, \boldsymbol{\beta}) \neq (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*) \right\} \right],$$

where $\boldsymbol{\alpha}^* \in \mathbb{R}^q$ and $\|\boldsymbol{\alpha}^*\| = 1$. $V_n(\boldsymbol{\alpha}, \boldsymbol{\beta})$ is the empirical version of $I(\mathbf{x}^\top \boldsymbol{\beta}, \mathbf{y}^\top \boldsymbol{\alpha})$, which is defined as

$$V_n(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^{n} \log \left\{ \frac{\widehat{f}_{\mathbf{x}^\top \boldsymbol{\beta}, \mathbf{y}^\top \boldsymbol{\alpha}}(\mathbf{x}_i^\top \boldsymbol{\beta}, \mathbf{y}_i^\top \boldsymbol{\alpha})}{\widehat{f}_{\mathbf{x}^\top \boldsymbol{\beta}}(\mathbf{x}_i^\top \boldsymbol{\beta}) f_{\mathbf{y}^\top \boldsymbol{\alpha}}(\mathbf{y}_i^\top \boldsymbol{\alpha})} \right\},$$

To solve for the $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, we apply Algorithm 1 repeatedly.

# Extension to Arbitrary Dimensions: Computation

1. Input error control $\epsilon$, maximum iteration $M$. Initialize $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ with a root-$n$ consistent estimator.
2. 2.1 Apply Step 2 in Algorithm 1 to the data $\{\mathbf{x}_i, \mathbf{y}_i^\top \boldsymbol{\alpha}_{m-1}\}$, $i = 1, \ldots, n$ to obtain $\boldsymbol{\beta}_m$.
   2.2 Apply Step 2 in Algorithm 1 to the data $\{\mathbf{y}_i, \mathbf{x}_i^\top \boldsymbol{\beta}_m\}$, $i = 1, \ldots, n$ to obtain $\boldsymbol{\alpha}_m$.
   2.3 Break if $\|\boldsymbol{\beta}_m - \boldsymbol{\beta}_{m-1}\|_2 + \|\boldsymbol{\alpha}_m - \boldsymbol{\alpha}_{m-1}\|_2 \leq \epsilon$.
3. Output $(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = (\boldsymbol{\alpha}_m, \boldsymbol{\beta}_m)$ if $V_n(\boldsymbol{\alpha}_m, \boldsymbol{\beta}_m) - c_n I\left\{(\boldsymbol{\alpha}_m, \boldsymbol{\beta}_m) \neq (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)\right\} > V_n(\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)$ and $(\widehat{\boldsymbol{\alpha}}, \widehat{\boldsymbol{\beta}}) = (\boldsymbol{\alpha}^*, \boldsymbol{\beta}^*)$ otherwise.

# Numerical Validation

Let $\mathbf{x} = (X_1, \ldots, X_p)^\top \sim N(0, \mathbf{I}_p)$, $\varepsilon \sim N(0, 1)$ is an independent error term and $a$ increases from 0 to 1. We also set $\boldsymbol{\beta}_1 = (1, 1, 0, \ldots, 0)^\top$ and $\boldsymbol{\beta}_2 = (1, 0, \ldots, 0)^\top$ with dimension $p = 10$. In each sub scenarios, the sample size vary from 50 to 100, respectively.

$$\begin{aligned}
&\text{Scenario 1.} && Y = a\mathbf{x}^\top \boldsymbol{\beta}_1 + \varepsilon; \\
&\text{Scenario 2.} && Y = a(\mathbf{x}^\top \boldsymbol{\beta}_1)^2 + \varepsilon; \\
&\text{Scenario 3.} && Y = a(\mathbf{x}^\top \boldsymbol{\beta}_1)(\mathbf{x}^\top \boldsymbol{\beta}_2) + 0.5\varepsilon.
\end{aligned}$$

The intensity parameter $a$ controls whether each model is under the null or the alternative hypothesis.

# Numerical Validation: Details

We implement three competitors in R:

- Mutual information test by Berrett and Samworth [2019]: "MINTperm" function from the "IndepTest" package.
- Distance correlation by Székely and Rizzo [2013]: "dcor.ttest" function from the "energy" package
- Ranks of distance by Heller et al. [2012]: "hhg.test" function from the "HHG" package.

We use 1000 permutations in each settings to estimate the $p$-value and repeat each setting 1000 times. The size and power of the tests are reported at two significance levels, i.e., $\alpha = 0.05$ and 0.1.

# Numerical results: Scenario 1

| $n$ | $\alpha$ | Test | $a = 0$ | $a = 0.2$ | $a = 0.4$ | $a = 0.6$ | $a = 0.8$ | $a = 1$ |
|---|---|---|---|---|---|---|---|---|
| 50 | 0.05 | MI | 0.059 | 0.110 | 0.283 | 0.562 | 0.852 | 0.990 |
| | | BS | 0.047 | 0.079 | 0.218 | 0.482 | 0.645 | 0.785 |
| | | DC | 0.065 | 0.198 | 0.608 | 0.910 | 0.988 | 0.996 |
| | | HG | 0.042 | 0.079 | 0.176 | 0.467 | 0.648 | 0.810 |
| | 0.1 | MI | 0.119 | 0.171 | 0.360 | 0.652 | 0.894 | 0.993 |
| | | BS | 0.098 | 0.153 | 0.330 | 0.620 | 0.757 | 0.876 |
| | | DC | 0.110 | 0.267 | 0.708 | 0.946 | 0.995 | 0.999 |
| | | HG | 0.085 | 0.146 | 0.295 | 0.601 | 0.751 | 0.881 |
| 100 | 0.05 | MI | 0.060 | 0.109 | 0.425 | 0.907 | 0.997 | 1.000 |
| | | BS | 0.046 | 0.101 | 0.501 | 0.865 | 0.978 | 0.998 |
| | | DC | 0.062 | 0.393 | 0.947 | 1.000 | 1.000 | 1.000 |
| | | HG | 0.040 | 0.095 | 0.471 | 0.832 | 0.961 | 0.993 |
| | 0.1 | MI | 0.110 | 0.184 | 0.523 | 0.941 | 0.998 | 1.0000 |
| | | BS | 0.111 | 0.209 | 0.644 | 0.935 | 0.989 | 0.998 |
| | | DC | 0.100 | 0.468 | 0.970 | 1.000 | 1.000 | 1.000 |
| | | HG | 0.085 | 0.182 | 0.607 | 0.908 | 0.981 | 0.995 |

# Numerical results: Scenario 2

| $n$ | $\alpha$ | Test | $a = 0$ | $a = 0.2$ | $a = 0.4$ | $a = 0.6$ | $a = 0.8$ | $a = 1$ |
|---|---|---|---|---|---|---|---|---|
| 50 | 0.05 | MI | 0.071 | 0.163 | 0.454 | 0.710 | 0.848 | 0.920 |
| | | BS | 0.047 | 0.128 | 0.260 | 0.368 | 0.438 | 0.465 |
| | | DC | 0.065 | 0.086 | 0.126 | 0.136 | 0.154 | 0.155 |
| | | HG | 0.042 | 0.137 | 0.215 | 0.316 | 0.328 | 0.349 |
| | 0.1 | MI | 0.114 | 0.220 | 0.500 | 0.744 | 0.861 | 0.929 |
| | | BS | 0.098 | 0.240 | 0.388 | 0.541 | 0.576 | 0.619 |
| | | DC | 0.110 | 0.144 | 0.179 | 0.206 | 0.214 | 0.221 |
| | | HG | 0.085 | 0.205 | 0.311 | 0.429 | 0.444 | 0.479 |
| 100 | 0.05 | MI | 0.064 | 0.240 | 0.870 | 0.998 | 1.000 | 1.000 |
| | | BS | 0.046 | 0.198 | 0.486 | 0.643 | 0.718 | 0.770 |
| | | DC | 0.062 | 0.094 | 0.157 | 0.207 | 0.254 | 0.269 |
| | | HG | 0.040 | 0.175 | 0.417 | 0.559 | 0.640 | 0.668 |
| | 0.1 | MI | 0.112 | 0.290 | 0.895 | 0.999 | 1.000 | 1.000 |
| | | BS | 0.111 | 0.313 | 0.643 | 0.787 | 0.830 | 0.873 |
| | | DC | 0.100 | 0.148 | 0.224 | 0.307 | 0.359 | 0.353 |
| | | HG | 0.085 | 0.268 | 0.546 | 0.685 | 0.737 | 0.773 |

# Numerical results: Scenario 3

| $n$ | $\alpha$ | Test | $a = 0$ | $a = 0.2$ | $a = 0.4$ | $a = 0.6$ | $a = 0.8$ | $a = 1$ |
|---|---|---|---|---|---|---|---|---|
| 50 | 0.05 | MI | 0.059 | 0.185 | 0.557 | 0.775 | 0.888 | 0.920 |
| | | BS | 0.039 | 0.152 | 0.311 | 0.419 | 0.454 | 0.513 |
| | | DC | 0.065 | 0.086 | 0.117 | 0.148 | 0.155 | 0.165 |
| | | HG | 0.042 | 0.125 | 0.227 | 0.321 | 0.357 | 0.369 |
| | 0.1 | MI | 0.120 | 0.248 | 0.602 | 0.797 | 0.909 | 0.932 |
| | | BS | 0.094 | 0.232 | 0.441 | 0.569 | 0.633 | 0.662 |
| | | DC | 0.110 | 0.149 | 0.180 | 0.218 | 0.219 | 0.235 |
| | | HG | 0.085 | 0.208 | 0.339 | 0.432 | 0.461 | 0.479 |
| 100 | 0.05 | MI | 0.061 | 0.356 | 0.948 | 0.998 | 1.000 | 1.000 |
| | | BS | 0.051 | 0.289 | 0.668 | 0.815 | 0.861 | 0.860 |
| | | DC | 0.062 | 0.095 | 0.179 | 0.209 | 0.253 | 0.270 |
| | | HG | 0.040 | 0.220 | 0.442 | 0.582 | 0.656 | 0.688 |
| | 0.1 | MI | 0.112 | 0.409 | 0.953 | 0.998 | 1.000 | 1.000 |
| | | BS | 0.102 | 0.433 | 0.791 | 0.905 | 0.932 | 0.926 |
| | | DC | 0.100 | 0.159 | 0.266 | 0.302 | 0.341 | 0.363 |
| | | HG | 0.085 | 0.307 | 0.573 | 0.694 | 0.771 | 0.793 |

# Numerical Validation

Next, we investigate the dimension effect for both $\mathbf{x}$ and $\mathbf{y}$. Suppose we draw $\mathbf{x} = (X_1, \ldots, X_p)^\top$ from a multivariate normal distribution with mean zero and covariance matrix $\mathbf{I}_p$, the error term $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_q)^\top \sim N(0, \mathbf{I}_q)$. We generate $\mathbf{y} = (Y_1, \ldots, Y_q)^\top$ as follows. Let $Y_1 = a\cos(\mathbf{x}^\top \boldsymbol{\beta}_1) + 0.5\exp(-a|\mathbf{x}^\top \boldsymbol{\beta}_1|)\varepsilon_1$, and $Y_j = 0.5\varepsilon_j$, for $j = 2, \ldots, q$, where $\boldsymbol{\beta}_1 = (1, 1, 0, \ldots, 0)^\top$. We let $p$ and $q$ vary among 2, 5 and 10 and study the performances of our test under both null and alternative hypothesises by letting $a = 0$ and $a = 1$, respectively.

# Numerical results: Scenario 4

| $a$ | $\alpha$ | Test | $p = 2$ | | | $p = 5$ | | | $p = 10$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $q = 2$ | $q = 5$ | $q = 10$ | $q = 2$ | $q = 5$ | $q = 10$ | $q = 2$ | $q = 5$ | $q = 10$ |
| 0 | 0.05 | MI | 0.064 | 0.070 | 0.061 | 0.067 | 0.069 | 0.069 | 0.066 | 0.060 | 0.060 |
| | | BS | 0.040 | 0.045 | 0.046 | 0.043 | 0.045 | 0.060 | 0.054 | 0.043 | 0.058 |
| | | DC | 0.062 | 0.080 | 0.063 | 0.065 | 0.060 | 0.056 | 0.056 | 0.069 | 0.067 |
| | | HG | 0.043 | 0.054 | 0.042 | 0.049 | 0.043 | 0.051 | 0.052 | 0.045 | 0.057 |
| | 0.1 | MI | 0.114 | 0.119 | 0.111 | 0.118 | 0.116 | 0.107 | 0.119 | 0.105 | 0.106 |
| | | BS | 0.093 | 0.102 | 0.102 | 0.096 | 0.095 | 0.102 | 0.112 | 0.089 | 0.110 |
| | | DC | 0.098 | 0.124 | 0.112 | 0.099 | 0.105 | 0.104 | 0.096 | 0.111 | 0.113 |
| | | HG | 0.086 | 0.093 | 0.086 | 0.087 | 0.097 | 0.103 | 0.093 | 0.100 | 0.105 |
| 1 | 0.05 | MI | 1.000 | 0.995 | 0.977 | 0.999 | 0.991 | 0.918 | 0.963 | 0.895 | 0.803 |
| | | BS | 0.730 | 0.467 | 0.305 | 0.288 | 0.201 | 0.130 | 0.144 | 0.105 | 0.099 |
| | | DC | 0.894 | 0.755 | 0.621 | 0.281 | 0.215 | 0.171 | 0.126 | 0.129 | 0.104 |
| | | HG | 0.997 | 0.938 | 0.827 | 0.614 | 0.490 | 0.349 | 0.319 | 0.237 | 0.169 |
| | 0.1 | MI | 1.000 | 0.997 | 0.990 | 1.000 | 0.992 | 0.940 | 0.973 | 0.925 | 0.845 |
| | | BS | 0.814 | 0.607 | 0.446 | 0.427 | 0.309 | 0.217 | 0.246 | 0.196 | 0.163 |
| | | DC | 0.942 | 0.843 | 0.718 | 0.372 | 0.307 | 0.260 | 0.199 | 0.183 | 0.171 |
| | | HG | 0.999 | 0.976 | 0.888 | 0.717 | 0.593 | 0.445 | 0.423 | 0.342 | 0.263 |

# A Real Data Example

In this section, we apply the proposed test to a stock portfolio performance data set [Liu and Yeh, 2017] from the UCI machine learning repository. The data contains 63 observations of quarterly recorded financial data which contains six predictors: the weight of large B/P concept, the weight of large ROE concept, the weight of large S/P concept, the weight of large return rate in the last quarter concept, the weight of large market value concept, and the weight of small systematic risk concept. We examine whether the absolute winning rate are independent of the six predictors.

# A Real Data Example

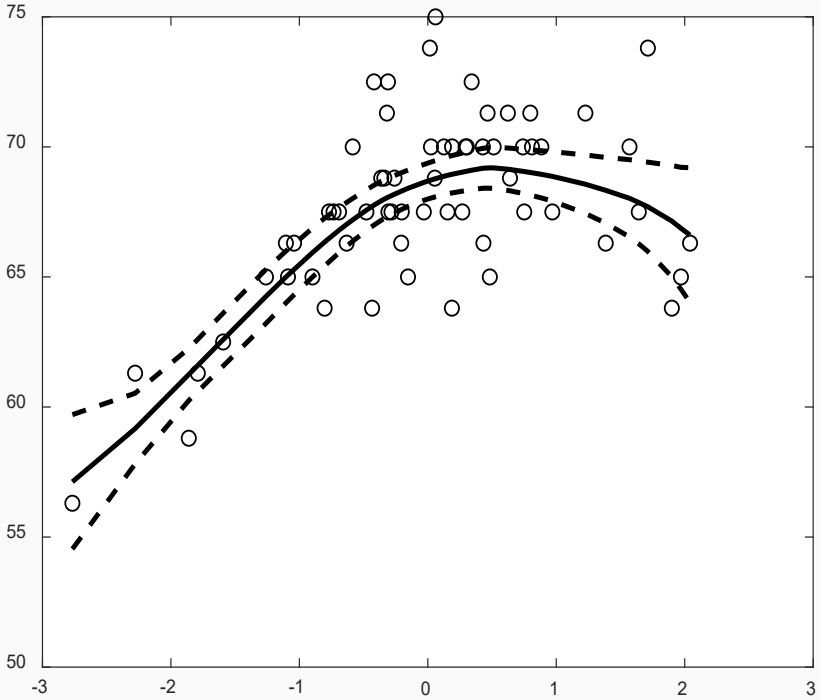
To make the problem more challenging, we add other six predictors whose entries are randomly and independently generated from $N(0, 1)$. We also summarize the corresponding $p$-values of these tests in Table 2.

# A Real Data Example

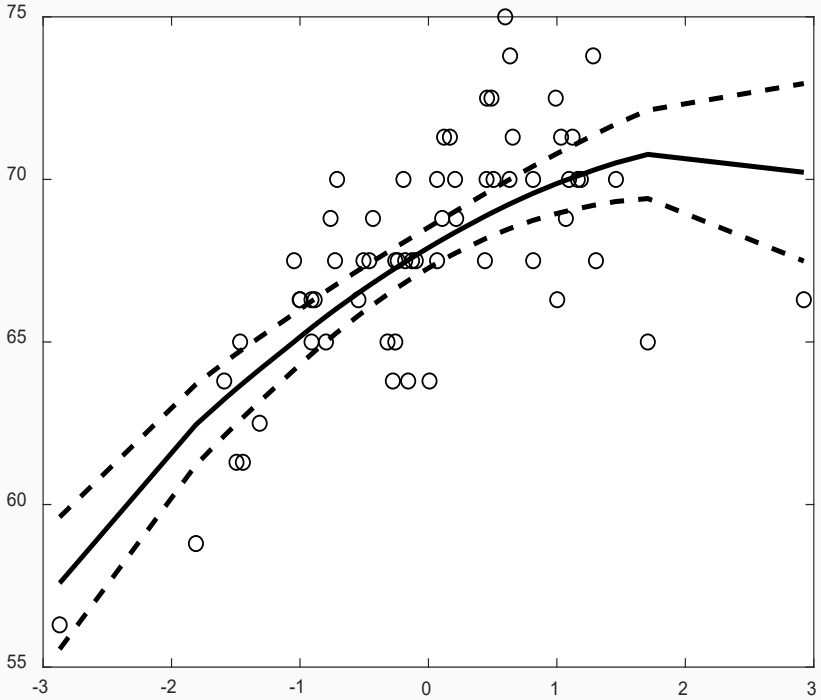
**Table 2:** $p$-value of the proposed test together with mutual information estimated by entropy (BS), distance correlation (DC), and ranks of distances (HG)

| Number of predictors | Test | MI | BS | DC | HG |
|---|---|---|---|---|---|
| 6 | $p$-value | $<0.001$ | 0.014 | $<0.001$ | 0.003 |
| 12 | $p$-value | $<0.001$ | 0.229 | 0.441 | 0.389 |

# A Real Data Example




(A)




(B)

**Figure 1:** The scatterplots of the absolute winning rate versus standardized $(\mathbf{x}_1^{\mathsf{T}}\widehat{\boldsymbol{\beta}}_1)$ in Panel (A) and versus standardized $(\mathbf{x}_2^{\mathsf{T}}\widehat{\boldsymbol{\beta}}_2)$ in Panel (B) in the Real Data Example.

# A Brief Summary

- By applying the idea of projection, we proposed a penalized mutual information to estimate the maximum unit direction consistently under both null and alternative hypothesis
- The proposed independence test is consistent against all fixed dependent alternatives. It can also detect local alternatives converging to the null at exact rate $n^{1/2} h_x^{1/4} h_y^{1/4}$
- The proposed test statistic behaves as if the model was univariate. The convergence rate of the statistic does not depend on the dimension of the $\mathbf{x}$.

**Questions?**

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